

Detecting Atrial Fibrillation (AFIB) Using an ECG Filtering Model

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Introduction

• Motivation

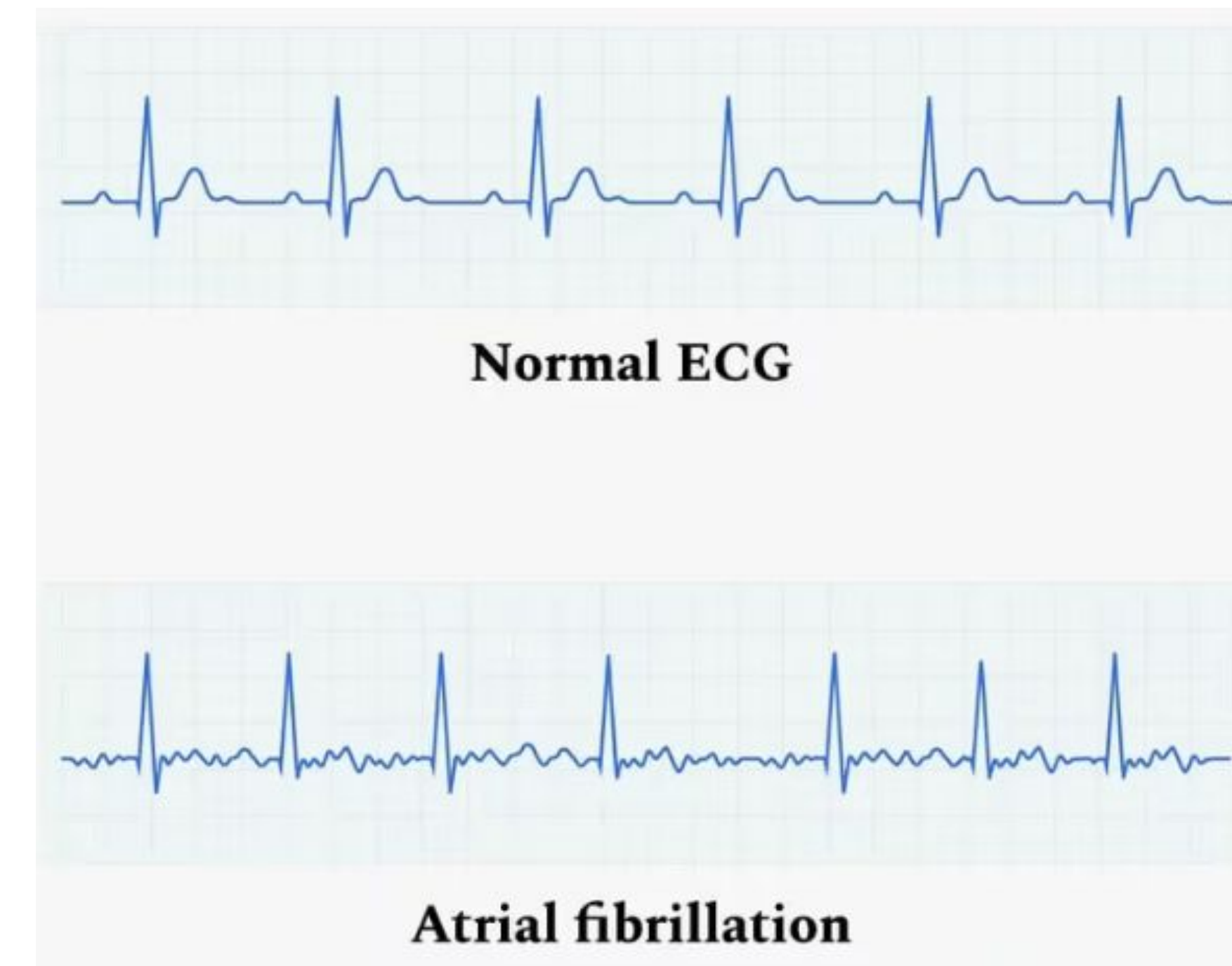
- Atrial Fibrillation (AFib) is a disease in which the atria of the heart beat erratically and out of sync with the ventricles. This disease can be characterized by a fast, irregular heart rate (1).
- AFib is a common condition that has seen an increase in numbers in recent years. Approximately 10.5 million individuals in the US have AFib.
- AFib “doubles the risk of mortality, is one of the most common causes of stroke, increases risks of heart failure, myocardial infarction, chronic kidney disease and dementia, and results in lower quality of life,” - Jean Jacques Noubiap, MD, PhD (2). An earlier diagnosis will allow patients to begin treatment sooner, improving their quality of life and decreasing the risk of other health implications.

• System used: MATLAB analysis of Electrocardiogram

- Input of system: Lead II of a 12-lead ecg plot
- Output of system: AFib detection

• R-R interval and P wave analysis correlation to AFib

- A typical R-R interval is around 0.6-1.1 seconds (4). AFib causes inconsistencies in the R-R intervals of an electrocardiogram (ECG), possibly creating an R-R interval over 1.1 seconds.
- In AFib, P waves are not usually present due to fibrillatory waves in between the QRS complexes.
- The system created will detect these inconsistencies and warn the user.



Approach/System

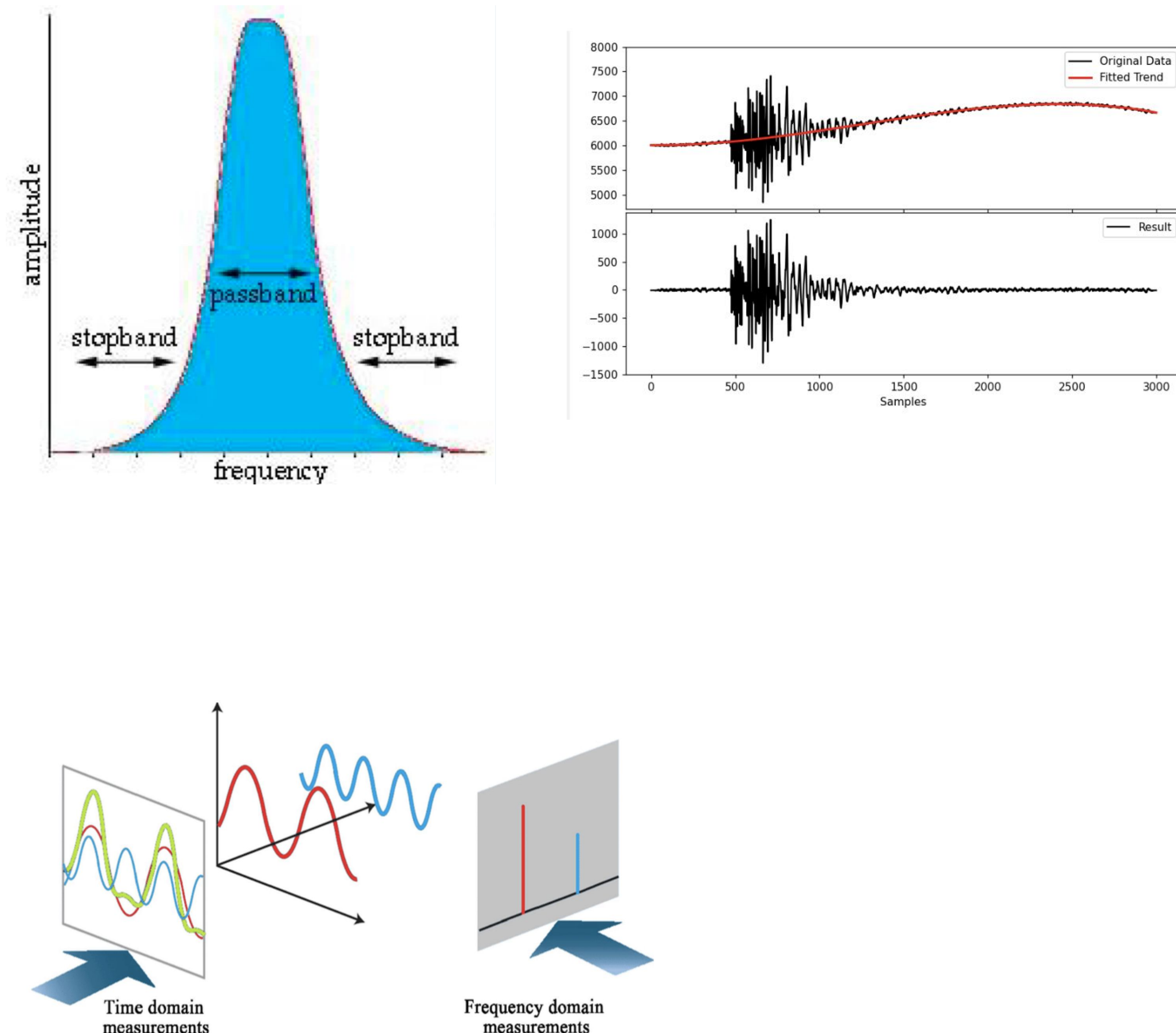
- **System:** MATLAB code was made to filter ECG data using a bandpass filter to minimize low/high frequency inputs, detrending to minimize baseline wander, and time domain analysis to correctly identify lengths of certain intervals (5).

- **Data:** Four 12-lead ECG plots.

- **Validation:** Manually calculating the R intervals and presence of the P wave and comparing that to the results generated by the system created.

- Lead II plots of each file were analyzed, due to the associations of the lead II plot and clear R-R intervals and P waves (3).

- The MATLAB code applies a bandpass filter and detrends the data to make the plot easier to analyze (4).
- R and P peaks are analyzed, and if the graph is inverted, it will flip it back to normal and analyze for these peaks. It will find the differences between R and P peaks and test for consistency.



Conclusion

- Lead II plots were successfully interpreted and analyzed using filtering techniques and by transferring ECG data into the time domain.
- Treatments tend to be successful when started early, making it important to find ways to detect AFib in its early stages.
- Irregular R-R intervals and the absence of P waves does correlate to AFib in patients.

Objectives/Aim

- **Objective:** To properly analyze electrocardiograms for AFib and warn the user if they are experiencing signs of AFib by developing a MATLAB code that can accurately evaluate ECG signals indicative of AFib through analysis of R-R intervals and P waves.

- Many do not catch signals of AFib before it turns into something fatal, such as heart failure or myocardial infarction.
- Being able to warn patients experiencing AFib before experiencing fatal consequences can lead to lower fatality rates as a result of AFib.



Results

The axes show that these graphs are in the time domain (mV vs. s). The y axis is being multiplied by an arbitrary unit of value 200 and the x axis is being multiplied by the frequency (500 samples/sec). To find the true time and voltage, divide the values found along the y axis by 200 and the x axis by 500.

Note: P wave and R wave analysis did not occur after row 5300, as the pulse on leads of every sample is inaccurate past row 5300.

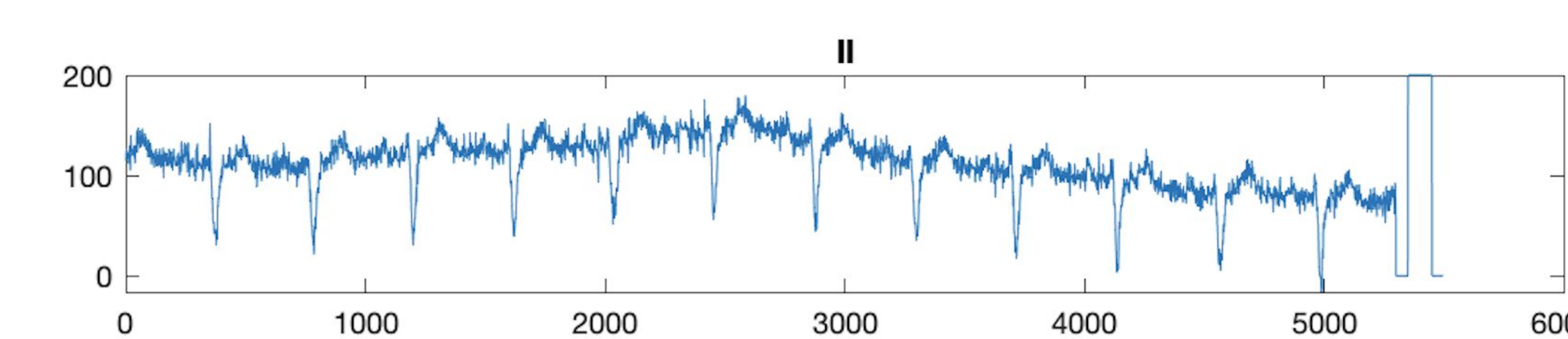


Figure 1. The original lead II plot of one of the 12-leads given for a patient experiencing AFib.

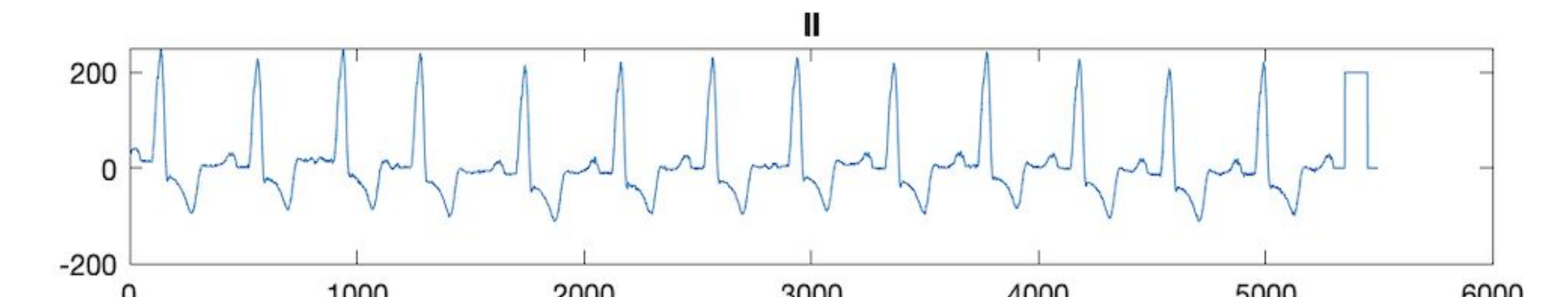


Figure 4. The original lead II plot of one of the 12-leads given for a patient not experiencing AFib.

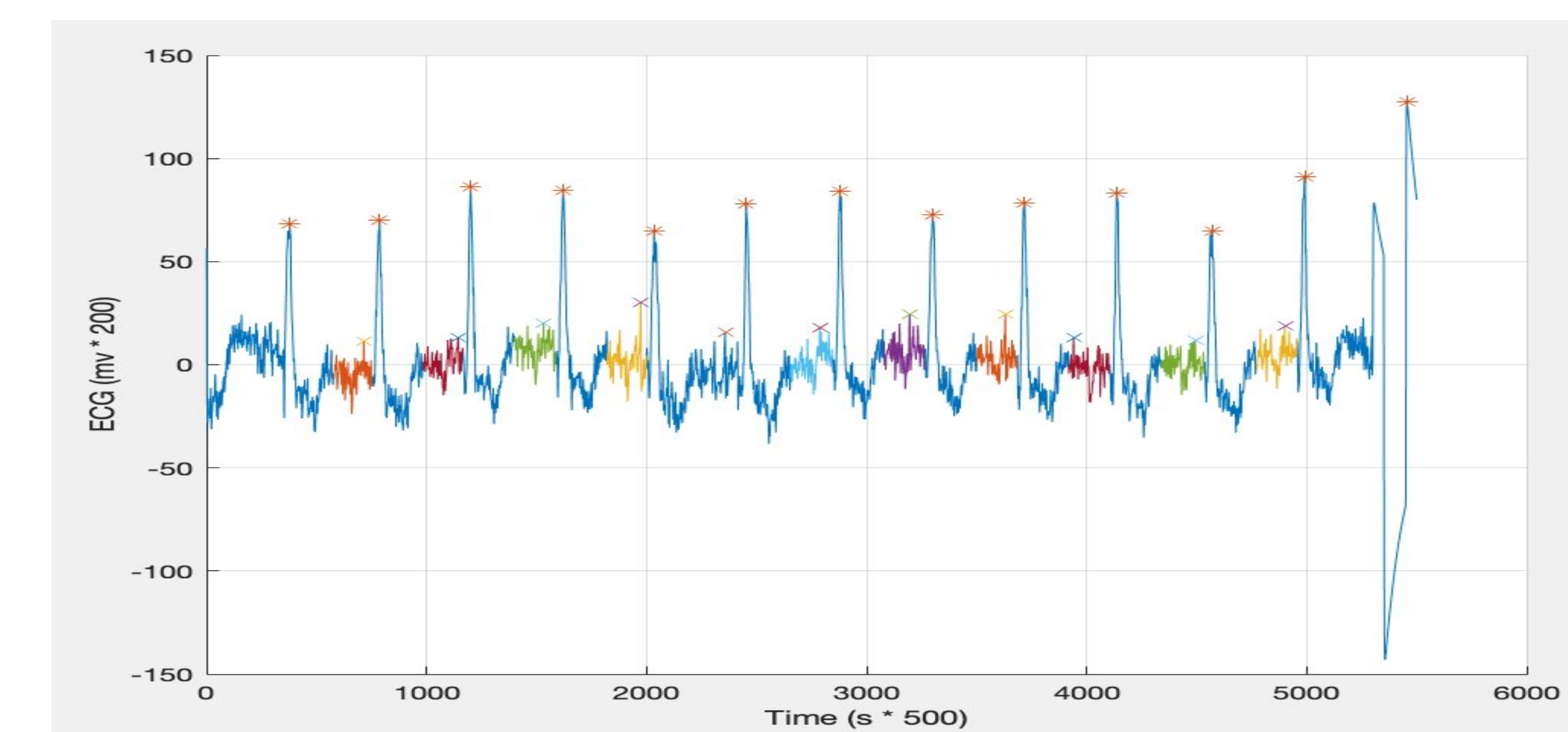


Figure 2. Application of a bandpass filter, detrending the data, and finding the R and P waves of each complex.

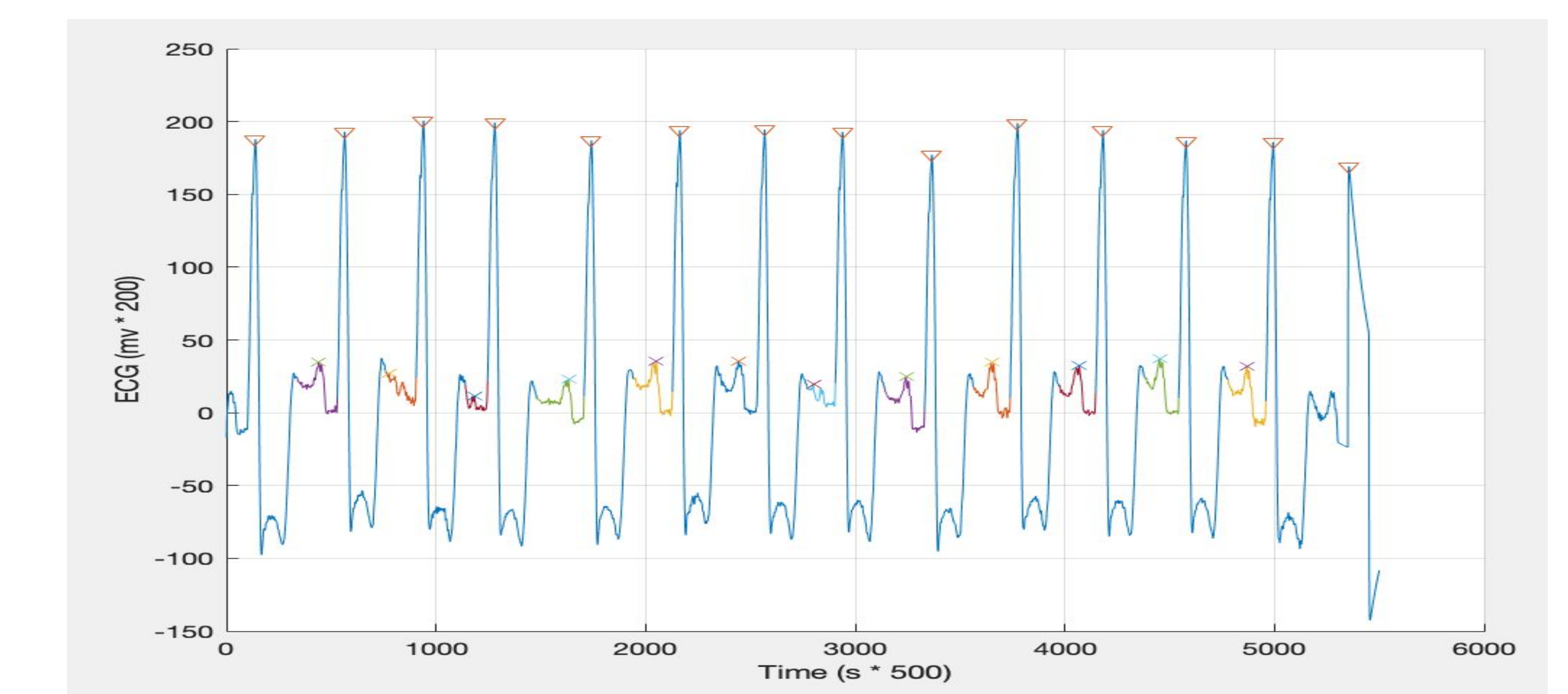


Figure 5. Application of a bandpass filter, detrending the data, and finding the R and P waves of each complex.

```
>> SystemsFinalCode
Consistent R intervals.
WARNING: Missing P wave. Seek medical attention.
```

Figure 3. The output of the code. It shows that all R intervals are consistent enough to not cause alarm, but the P intervals are inconsistent because of fibrillatory waves in between QRS complexes. This results in a warning message.

```
>> SystemsFinalCode
Consistent R intervals.
Consistent P waves.
```

Figure 6. Both R and P waves with consistent intervals; therefore, no warning messages occur.

References

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